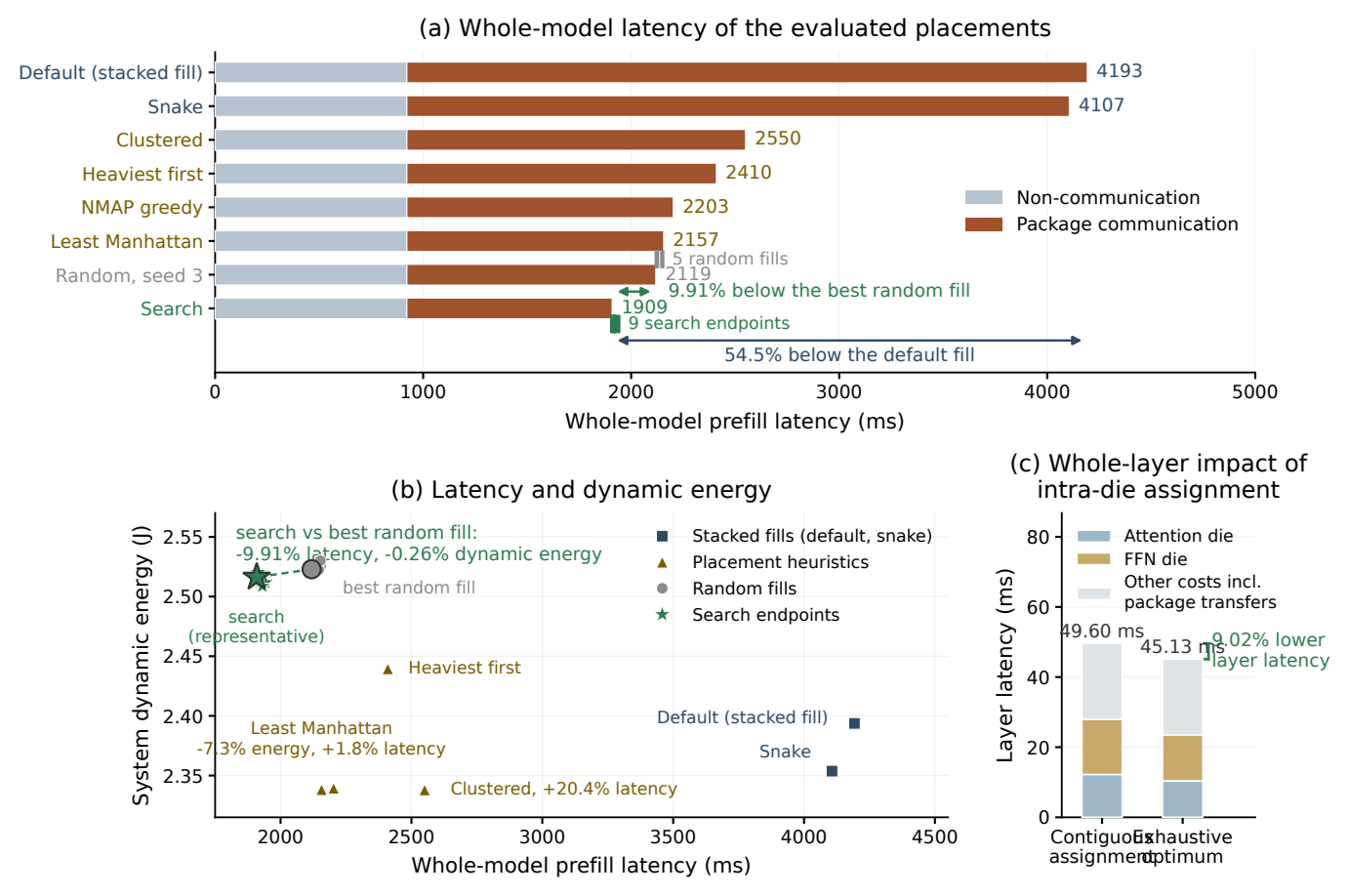
(a) Whole-model latency of the evaluated placements
Default (stacked fill)
Snake
Clustered
Heaviest first
NMAP greedy
Least Manhattan
Random, seed 3
Search
4193
4107
2550
2410
2203
2157
5 random fills
2119
9.91% below the best random fill
1909
9 search endpoints
54.5% below the default fill
Non-communication
Package communication
0
1000
2000
3000
4000
5000
Whole-model prefill latency (ms)
(b) Latency and dynamic energy
search vs best random fill:
-9.91% latency, -0.26% dynamic energy
best random fill
search
(representative)
Heaviest first
Least Manhattan
-7.3% energy, +1.8% latency
Clustered, +20.4% latency
Default (stacked fill)
Snake
Stacked fills (default, snake)
Placement heuristics
Random fills
Search endpoints
System dynamic energy (J)
2.55
2.50
2.45
2.40
2.35
2000
2500
3000
3500
4000
4500
Whole-model prefill latency (ms)
(c) Whole-layer impact of
intra-die assignment
Attention die
FFN die
Other costs incl.
package transfers
Layer latency (ms)
80
60
40
20
0
49.60 ms
45.13 ms
9.02% lower
layer latency
Contiguous
assignment
Exhaustive
optimum